

NITYA MITTAL

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WORK EXPERIENCE

- American Express** | *Software Engineer - Associate DPM* July 2023 - July 2024
- **Saved ~\$70K** in Q1–Q2'24 through **33% data rationalization**, by designing & developing **Python** and **SQL** pipelines to resolve data quality issues and implementing optimization strategies for the **Hive** data layers used by the Anti-Money Laundering Transaction Monitoring team, Global Risk & Compliance Dept.
 - Led the implementation of onboarding pipelines for the Kabbage acquisition, migrating **~20K accounts** into Amex's banking ecosystem through scalable ingestion, validation, and reconciliation frameworks that ensured AML compliance.
 - Maintained and debugged production pipelines ingesting **millions of transactions** per 5-minute Amex feed, detecting file loss and duplication through transaction-volume anomaly monitoring in **Yellowbrick**; partnered with the AML Transaction Monitoring team via **JIRA** in an Agile environment to triage and resolve data-integrity issues.
- American Express** | *Data Engineering - Analyst Intern* May 2022 - July 2022
- Reduced development time by **80%** for Model Performance Tracking of the Expected Credit Loss Forecasting Tool, by building **Python** scripts in **MLStudio** to consolidate all US and global portfolios at the Credit & Fraud Risk Dept.
 - Automated data sourcing across multiple formats (SQL, JSON) by designing object-oriented **Java** services with secure coding principles to pull and load data into **Hive**, following SDLC.
 - Streamlined onboarding for new data engineers by authoring technical design documentation in **Confluence** detailing BAU and ad-hoc workflows, sharing knowledge to support the team.

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB, PHP, R, SQL, PostgreSQL, HiveQL, TypeScript, JavaScript
Cloud Tech: Amazon Web Services: EC2, S3, Lambda, ECS, CodePipeline; Google Cloud: GCP, Vertex AI, Firestore.
DevOps, Infra & Delivery: Git, Docker, Kubernetes, Apache Kafka, Redis, Linux OS, Slurm, SDLC, CI/CD, Jenkins.
App Dev & Analytics: HTML, CSS, REST APIs, React, Node.js, FastAPI, Uvicorn, Streamlit, n8n, PHPUnit.
ML & AI: scikit-learn, numpy, scipy, pandas, LangChain, keras, tensorflow, LangGraph, transformers; BERT, Gemini 2, RoBERTa, all-MiniLM-L6-v2, all-mpnet-base-v2, VGG19, Opacus, Presidio, GPT-2, NetworkX.
AI Coding Tools: Claude Code, Codex, Cursor, GitHub Copilot

EDUCATION

- University of Michigan, Ann Arbor** Aug. 2024 – May 2026
Master of Science in Engineering in Computer Science **GPA: 4.0/4.0**
- Delhi Technological University, Delhi** July 2019 – May 2023
Bachelor of Technology in Mathematics & Computer Science **GPA: 3.81/4.0**

Coursework: Data Structures, Algorithms, Object Oriented Programming, Software Engineering, System Design, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, Natural Language Processing, Data Science, Statistics, Linear Algebra.

PUBLICATION

“**University Recommendation System**” presented at **International Research Conference on Science Technology, Engineering and Management (IRCSTEM)** on 6–7th May 2023 at Gurgaon, India.

PROJECT EXPERIENCE

- Wikispeech (MediaWiki extension): Open Source Contribution** Jan 2026 – Jun 2026
- Built Wikimedia's article-to-audio download feature, enabling users to listen to and **save Wikipedia articles as speech**; delivered 4 **PHP** patches spanning schema design, **REST APIs**, JobQueue processing, and UI workflows, backed by 40+ **PHPUnit** tests.
- CineRank: Real-Time Movie Recommendation System** Aug 2025 – Oct 2025
- Built a real-time recommendation engine that continuously updates user preferences & serves movie suggestions in **<100ms latency** on the **MovieLens 32M** dataset. Designed a streaming ML pipeline using **Kafka**, **Redis**, **PostgreSQL**, & **FastAPI**, combining SVD collaborative filtering, cold-start fallback logic, & cached recommendation retrieval.
- Needle: RAG-based Research Assistant** Aug 2025 – Nov 2025
- Built a research assistant for literature discovery & source-grounded Q&A over **1.7M+ arXiv papers** & user-uploaded PDFs, using retrieval-augmented generation to ground answers in retrieved sources. Engineered the pipeline on **Vertex AI** (Gemini 2.0 Flash + text-embedding-004), with **Vertex Vector Search** for semantic retrieval, **Firestore** for chunk metadata, and a **Streamlit** UI; integrated OpenCitations/Crossref to surface per-paper citation counts.
- PrivNLGBench: Differential Privacy Benchmark** Jan 2025 – Jun 2025
- Built a benchmark for evaluating privacy-preserving NLP systems by transforming real-world datasets (Yelp, Enron, Clinton FOIA) into structured test sets with isolated PII and PII-free summaries. Developed the evaluation pipeline using **PyTorch**, **Opacus**, **DistilRoBERTa**, & **NetworkX**, achieving **0.85 accuracy** / **0.71 macro-F1** for PII detection and improving calibration from **0.272 to 0.083 ECE**.
- MiniGPT: From-Scratch GPT-Style Language Model** Sep 2025 – Oct 2025

- Implemented a decoder-only GPT-style **transformer** from scratch in **PyTorch** (multi-head causal self-attention, positional embeddings, & residual blocks), pretrained on **~1B tokens** of FineWeb-Edu via next-token prediction; applied supervised fine-tuning on SmolTalk multi-turn dialogues with assistant-only loss masking, then evaluated on a multiple-choice QA benchmark.

AI-Generated Text Detector: AI-Written Text Classifier

Aug 2025 - Oct 2025

- Built an AI-written-text classifier reaching **~80% accuracy** with a **scikit-learn/scipy** logistic-regression model over sparse features, and benchmarked SGD, RMSprop, & AdamW optimizers on a **PyTorch** variant.

ACHIEVEMENTS, EXTRACURRICULARS & LEADERSHIP

- **Awards:** McKinsey Next Generation Women Leaders 2023 & 2025 | URAM (Micron) Scholarship 2022 | AWS ML Scholarship 2021 | UCMAS International Distinction Grade 10
- **Leadership:** Judge, MHacks 17 (MLH Official) | Vice Chairperson, IEEE DTU CS | Student Advisory Board, Michigan Engineering | Networking Officer, GradSWE
- **Mentorship:** Mentored 300+ students at IEEE DTU, JA India & American Express, Joy of Coding (UMich)